## Module:

In [ ]:
```
It is a python file with .py extension conatining python
defination and statement and function and variable and classes.
```

## Package

In [ ]:
```
Collection of varius modules with path attribute.
```

## Library

In [ ]:
```
Collection of modules anf packages
```

In [2]:
```
import numpy as np
import pandas as pd
import seaborn as sns
```

## Imporatant keyword

In [ ]:
```
Import>> It is used to import modules , library and pachages.
as>> It used to rename modules.
from>> Import specific attributes (variable,function and classes)
```

## OS Module

In [ ]:
```
os>>This module is designed for interacting with operating systems.

os is standard utility module in python
```

In [3]:
```
import os
import shutil
import glob
```

# OS Module Functions:

In [ ]:
```
OS Module:
    1)os.rename()
    2)os.remove()
    3)os.path.exists()
    4)os.mkdir()
    5)os.makedirs()
    6)os.rmdir()
    7)os.listdir()
    8)os.getcwd()
    9)os.path.join()

shutil Module:
    1)shutil.rmtree()
    2)shutil.copy()

Glob Module:
    1)glob.glob(folder_path+"*.txt")
```

## 1.Rename

In [ ]:
```
Syntax:
    os.rename(source_file_path,destination_file_path)
```

In [5]:
```
os.rename('DL123.txt','DeepLearning111.txt')
```

## 2.Remove Files

In [ ]:
```
Syntax:os.remove(file_path)
```

In [11]:
```
os.remove("DeepLearning111.txt")
```

In [12]:
```
file_path="G:\Techno Savvy Institude\Informatica Tool\Python25-12-25\sample.txt
os.remove(file_path)
```

## 3.Check file exist or not

In [ ]:
```
os.path.exists(file_path)

os.path.exists(folder_path)
```

In [13]:
```
os.path.exists("DeepLearning111.txt")
```

Out[13]: False

In [15]:
```
os.path.exists("Deep_learning.txt")
```

Out[15]:
```
True
```

In [17]:
```
model_file_path="random_forest_model.pkl"

if os.path.exists(model_file_path):
    os.remove(model_file_path)
    print("File remove")
else:
    print("File not found")
```

```
File remove
```

In [19]:
```
folder_path="Movie"

if os.path.exists(folder_path):
    print("Folder is present")
```

```
Folder is present
```

## 4.Create Folder

In [ ]:
```
os.mkdir(folder_path)
```

In [24]:
```
os.mkdir('Images1')
```

In [25]:
```
os.mkdir('Images1')
```

```
---------------------------------------------------------------------------
FileExistsError                           Traceback (most recent call last)
Cell In[25], line 1
----> 1 os.mkdir('Images1')

FileExistsError: [WinError 183] Cannot create a file when that file already ex
ists: 'Images1'
```

In [44]:
```
folder_path="Mycode"

if os.path.exists(folder_path):
    print("Folder already present in directory")
else:
    os.mkdir(folder_path)
    print("Suceesfully created folder")
```

```
Suceesfully created folder
```

In [27]:
```
folder_path="Mycode"

if os.path.exists(folder_path):
    print("Folder already present in directory")
else:
    os.mkdir(folder_path)
    print("Suceesfully created folder")
```

```
Folder already present in directory
```

## 5.os.makedirs()

In [41]:
```
ML_algo_folder_path=r'Data_science\ML\Supervised_ML\Classification'

if not os.path.exists(ML_algo_folder_path):
    os.makedirs(ML_algo_folder_path)
    print("Folder created")
```

```
Folder created
```

## 6.Delete Empty Directory

In [ ]:
```
os.rmdir(folder_path)
```

In [29]:
```
os.rmdir('Movie')
```

In [31]:
```
folder_path='Movie'

if os.path.exists(folder_path):
    os.rmdir('Movie')
    print("Folder deleted")
else:
    print("Folder not found")
```

```
Folder not found
```

## 7.Delete Non empty Directory

In [33]:
```
import shutil
```

In [ ]:
```
shutil.rmtree(folder_path)
```

In [42]:
```
folder_path='Data_science\ML\Supervised_ML\Classification'

if os.path.exists(folder_path):
    shutil.rmtree(folder_path)
    print("Folder deleted")
```

```
Folder deleted
```

In [43]:
```
folder_path='Mycode'

if os.path.exists(folder_path):
    shutil.rmtree(folder_path)
    print("Folder deleted")
```

## Check File or Folders ion Directory

In [ ]:
```
os.listdir(folder_path)
```

In [46]:
```
os.listdir('Images1')
```

Out[46]:
```
['aadhar.jpg', 'Accuracy._md.png']
```

In [48]:
```
folder_path=r"C:\Users\Sai\TYBCS-2026\Java_Basics"
os.listdir(folder_path)
```

Out[48]:
```
['.classpath', '.project', '.settings', 'bin', 'src']
```

## 9.Current working directory

In [49]:
```
os.getcwd()
```

Out[49]:
```
'C:\\Users\\Sai'
```

## 10.Join Multiple Pth

In [ ]:
```
Synatx:
    os.path.join(folder_path1,folder_path2)
```

In [50]:
```
os.path.join('Database','Images1')
```

Out[50]:
```
'Database\\Images1'
```

In [53]:
```
folder_path1="Mycode"
folder_path2="myproject"
result=os.path.join(folder_path1,folder_path2)
result
```

Out[53]:
```
'Mycode\\myproject'
```

In [54]:
```
import glob
```

In [56]:
```
folder_path="myproject\\myproject\\"
file=glob.glob(folder_path+"*.py")
file
```

Out[56]:
```
['myproject\\myproject\\asgi.py',
 'myproject\\myproject\\settings.py',
 'myproject\\myproject\\urls.py',
 'myproject\\myproject\\wsgi.py',
 'myproject\\myproject\\__init__.py']
```

## Copy Files

In [ ]:
```
shutil.copy(source_file_path,destination_file_path)
```

In [57]:
```
destination_file_path="G:\Techno Savvy Institude\AI and DS"
source_file_path="Data12.txt"
shutil.copy(source_file_path,destination_file_path)
```

Out[57]:
```
'G:\\Techno Savvy Institude\\AI and DS\\Data12.txt'
```

In [ ]: